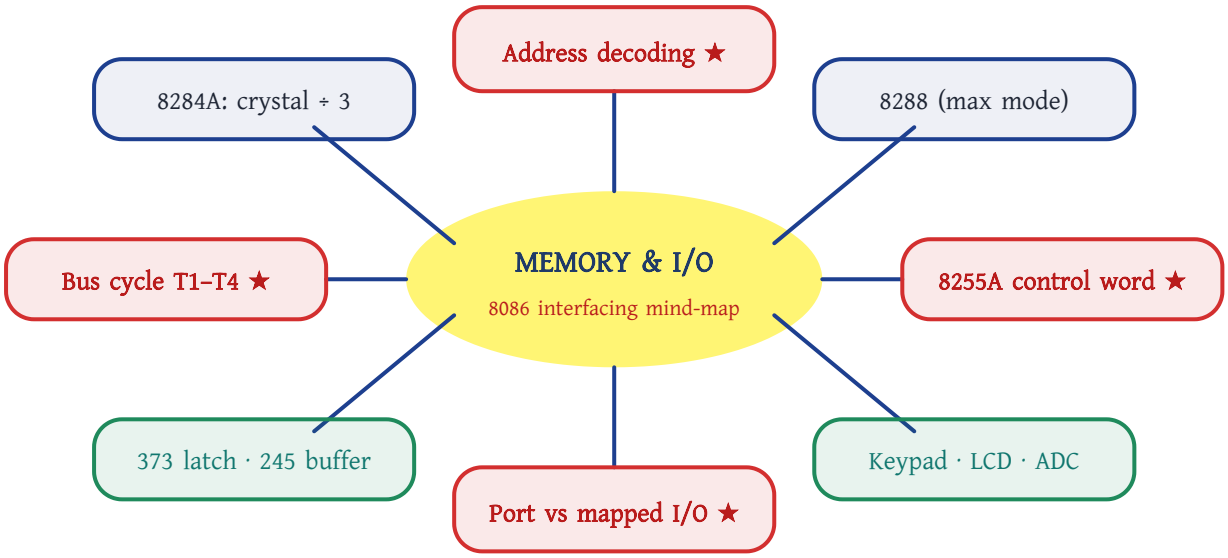
Memory & I/O Interfacing

8284A · 8288 · latches & transceivers · address decoding · 8255A PPI: exam-ready in 11 sheets



↑ the whole lesson on one page: red = high-yield, green = hardware / applied

Syllabus coverage

"**Clock and bus controller interfacing**: clock generator, bus demultiplexer, bus controller interfacing; **Memory Interfacing**: SRAM and EEPROM interfacing, Types of I/O: parallel I/O, programmed I/O, interrupt driven I/O, **I/O port address decoding**, **programmable peripheral interface (8255A)**, interface examples: Keyboard matrix, LCD/7-Segment display, printer, stepper motor, A/D and D/A converter". Pins: lesson mp1 · 8254/8251: mp3 · 8259A, DMA: mp4.

High-yield list

① CLK = crystal ÷ 3
② ALE → 74LS373
③ 2^n cells → n lines
④ 74LS138 memory map
⑤ IN/OUT: AL/AX, DX
⑥ 8255A: 80H, 9BH, BSR

Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example · $\overline{RD}$ = active-low signal

Contents

✓ Min/max mode, 8284A clock ..... p. 2
✓ Bus cycle, wait states, 8288 ..... p. 3
✓ Demultiplexing & buffering ..... p. 4
✓ Memory chips & banks ..... p. 5
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1 Why the 8086 Needs Support Chips

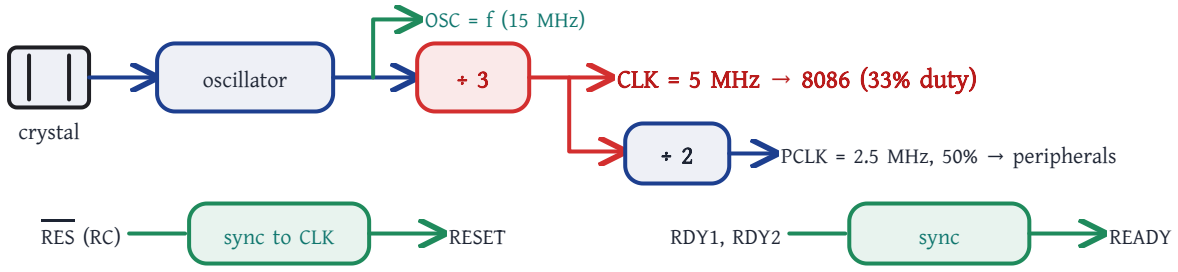
Idea: Address and data **share pins** (AD0–AD15, to fit 40 pins), each pin drives only **~10 74LS loads**, and there is no on-chip oscillator. So a board adds the **8284A** clock, **74LS373** latches, **74LS245** transceivers and, in maximum mode, the **8288** bus controller.

Pins	Minimum mode (MN/$\overline{MX}$ = 1)	Maximum mode (MN/$\overline{MX}$ = 0)
24–25	$\overline{INTA}$, ALE	QS1, QS0 (queue status)
26–28	$\overline{DEN}$, DT/$\overline{R}$, M/$\overline{IO}$	$\overline{S0}$, $\overline{S1}$, $\overline{S2}$ (status → 8288)
29–31	$\overline{WR}$, HLDA, HOLD	$\overline{LOCK}$, $\overline{RQ}$/$\overline{GT1}$, $\overline{RQ}$/$\overline{GT0}$
Use	single CPU makes its own controls	8087 / multiprocessor: **8288 makes them**

Unchanged in both modes: $\overline{RD}$, READY, RESET, CLK, $\overline{BHE}$/S7. On the 8088, pin 28 is IO/$\overline{M}$ (opposite sense).

2 8284A Clock Generator

Definition: An 18-pin chip that turns a crystal into **CLK**, **PCLK** and **OSC** and **synchronises READY and RESET** with the clock. CLK = $f_{crystal}$ ÷ 3 with a **33% duty cycle**.



one crystal, three clocks, plus clean RESET and READY for the CPU

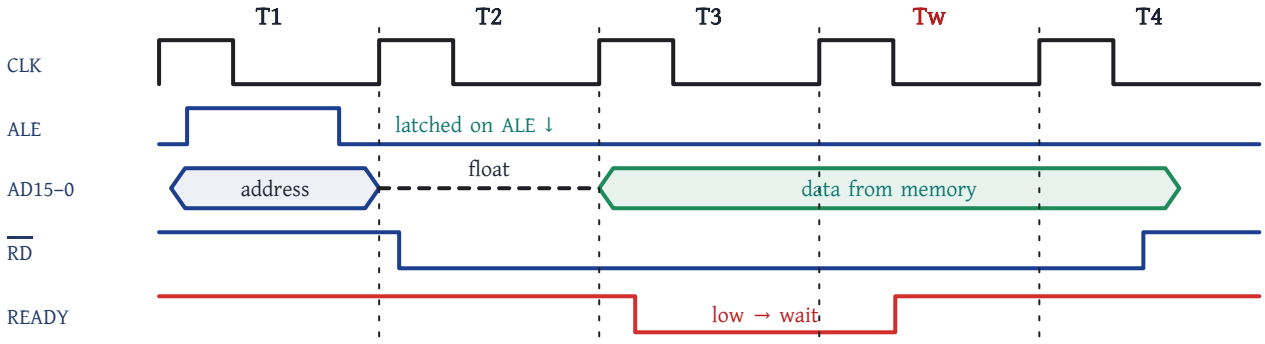
Worked example ✍️: 15 MHz crystal → CLK = **5 MHz** (T = 200 ns), PCLK = 2.5 MHz, OSC = 15 MHz. An 8086-2 (8 MHz) needs a **24 MHz** crystal; an 8086-1 (10 MHz) a 30 MHz one.

Memory trick: "**OSC ÷ 1, CLK ÷ 3, PCLK ÷ 6**" (all from the crystal). F/$\overline{C}$ = 0 uses the crystal, 1 the external EFI input.

Exam trap: CLK duty is **33%**; PCLK is the 50% one. RESET must stay HIGH ≥ **4 clocks**; then CS = FFFFH, IP = 0000H, so the first fetch is at **FFFF0H**: the boot EPROM sits at the top of memory (p. 6).

3 The Bus Cycle: T1–T4 and Wait States

Definition: One memory or I/O read/write = a **bus cycle** of **4 clocks**, T1–T4. A slow memory or port pulls **READY** low and the CPU inserts **wait states (Tw)**, one clock each.



a read cycle with one wait state (Intel's drawing puts Tw between T3 and T4)

State	What happens
T1	address on AD0–AD15, A16–A19, $\overline{\text{BHE}}$; **ALE pulses**; M/$\overline{\text{IO}}$, DT/$\overline{\text{R}}$ set
T2	address removed; $\overline{\text{RD}}$ or $\overline{\text{WR}}$ low; $\overline{\text{DEN}}$ low; write data out
T3	memory/port drives the data; READY is checked
Tw	repeated while READY = 0; one clock each; signals frozen
T4	CPU takes read data; $\overline{\text{RD}}$/$\overline{\text{WR}}$ go high; cycle ends

Worked example ✍️: 5 MHz: T = 200 ns, cycle = 4T = **800 ns**; 1 Tw → 1000 ns. Peak rate = 2 bytes / 800 ns = 2.5 MB/s.

Exam trap: Brey puts Tw between T2 and T3, Intel between T3 and T4; either way **READY low ⇒ +1 clock per Tw.**

4 8288 Bus Controller (Maximum Mode)

$\overline{\text{S2}}$ $\overline{\text{S1}}$ $\overline{\text{S0}}$	Cycle	Output
0 0 0	interrupt ack	$\overline{\text{INTA}}$
0 0 1	I/O read	$\overline{\text{IORC}}$
0 1 0	I/O write	$\overline{\text{IOWC}}$, $\overline{\text{AIOWC}}$
0 1 1	halt	none
1 0 0	code fetch	$\overline{\text{MRDC}}$
1 0 1	memory read	$\overline{\text{MRDC}}$
1 1 0	memory write	$\overline{\text{MWTC}}$, $\overline{\text{AMWC}}$
1 1 1	passive	none

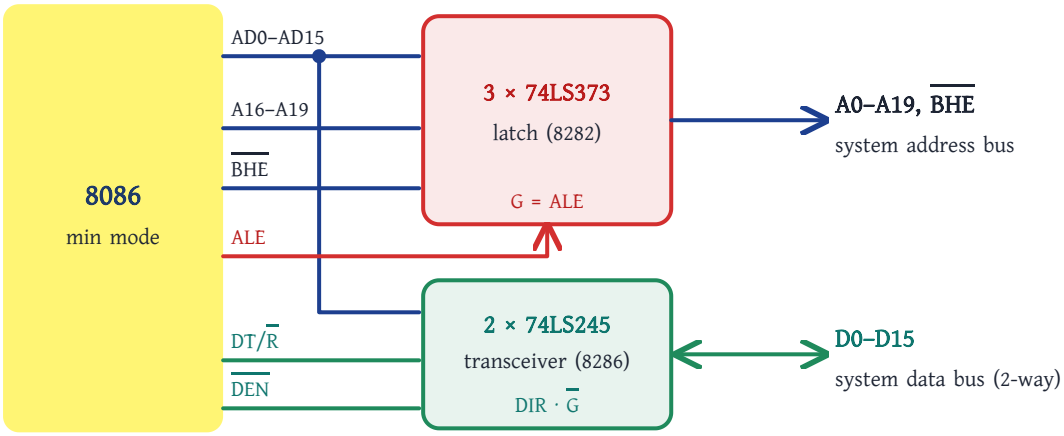
8288: 20 pins; turns the status bits into **commands** and ALE, DEN, DT/$\overline{\text{R}}$. "A" = **advanced** write, one clock early.

Memory trick: $\overline{\text{S2}}$ = 0 → **I/O side**, 1 → memory. Low bits: 00 special, 01 read, 10 write, 11 idle.

Exam trap: Bus arbitration is the **8289**'s job, not the 8288's. 8288 DEN is **active HIGH**.

5 Demultiplexing the Address/Data Bus

Definition: AD0–AD15 carry the **address in T1** and **data in T2–T4**. **ALE** (Address Latch Enable) is high in T1; a **latch** passes the address while ALE = 1 and **holds it from ALE's falling edge** to the end of the cycle. The latched lines form the system address bus A0–A19.



the classic buffered 8086: latch the address, buffer the data

Chip	What it is	Job in an 8086 system
74LS373 (8282)	octal **transparent latch**, 3-state	address latch: G ← ALE, $\overline{OE}$ ← GND
74LS374	octal **edge-triggered** D flip-flop	output port, clocked by $\overline{IOWC}$
74LS244	octal 3-state **buffer**, one way	address/control buffer, input port
74LS245 (8286)	octal **transceiver**, two way	data bus: DIR ← DT/$\overline{R}$, $\overline{G}$ ← $\overline{DEN}$

74LS245 truth:

$\overline{G}$ = 0, DIR = 1 → A → B (CPU **writes**)

$\overline{G}$ = 0, DIR = 0 → B → A (CPU **reads**)

$\overline{G}$ = 1 → both sides **Z** (isolated)

DT/$\overline{R}$: 1 = transmit, 0 = receive

Chip count ✍️:

8086: 21 lines to latch (AD0–15, A16–19, $\overline{BHE}$) → **3 × 373**; 16 data lines → 2 × 245

8088: only AD0–AD7 multiplexed → 2 × 373, A8–A15 via a 244, **1 × 245**

Memory trick: "**A**LE → **A**ddress latch; **D**EN → **D**ata enable; DT/$\overline{R}$ → **D**irection". 37**3** latch, 24**4** one-way, 24**5** both ways.

Exam trap: Buffers are for **fan-out** (> ~10 LS loads) and for floating the bus during DMA. The 16-bit 8086 needs **two** 245s; the 8088 one.

6 Memory Chips and Their Pins

Definition: An **N × M** chip has N locations (**$\log_2$N address lines**) of M bits (M data lines). Control pins: **$\overline{\text{CS}}$** chip select ← decoder, **$\overline{\text{OE}}$** ← $\overline{\text{MRDC}}$/$\overline{\text{RD}}$, **$\overline{\text{WE}}$** ← $\overline{\text{MWTC}}$/$\overline{\text{WR}}$.

Size	Lines
1K	10
2K	11
4K	12
8K	13
16K	14
32K	15
64K	16
1M	20

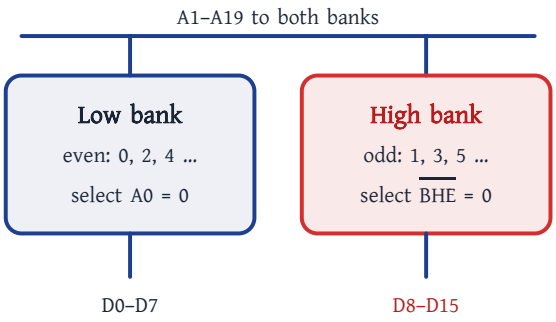
Chips	Type	Size (× 8)
6116 / 6264 / 62256	SRAM	2K / 8K / 32K
2716 / 2764 / 27256	EPROM	2K / 8K / 32K
2816 / 2864	EEPROM	2K / 8K

Memory trick: Digits after the family code = **Kbits**: 27**64** = 64 Kbit = **8 KB**; 62**256** = 32 KB; 61**16** = 2 KB.

Chips needed ✍️: $(N_{total} \div N_{chip}) \times (M_{total} \div M_{chip})$: 32K × 16 from 8K × 8 → 4 × 2 = **8 chips**.

	SRAM	EEPROM
Volatile?	yes	no (keeps data)
Write	tens of ns, unlimited	**≈ 1–10 ms**, ~10^5–10^6 cycles
Interface	A, D, $\overline{\text{CS}}$, $\overline{\text{OE}}$, $\overline{\text{WE}}$	same pins; wait or poll (D7) after a write
Used for	working memory, stack	settings, calibration data

7 8086 Even and Odd Memory Banks



1 MB = two 512 KB banks

$\overline{\text{BHE}}$	A0	Transfer
0	0	whole **word** (even address)
0	1	byte, **odd** bank (D8–D15)
1	0	byte, **even** bank (D0–D7)
1	1	none

Exam trap: A word at an **odd** address needs **2 bus cycles**. The 8088 has one bank and no $\overline{\text{BHE}}$.

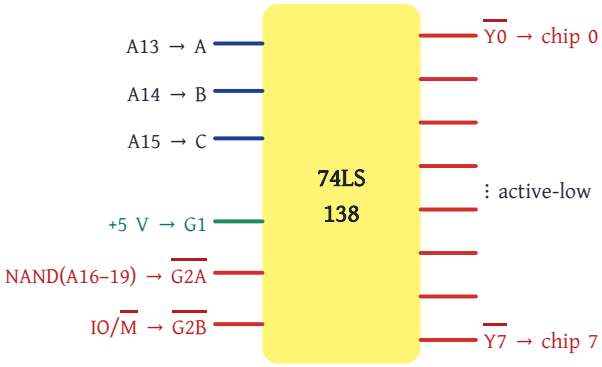
8 Address Decoding: Full vs Partial

Idea: The **low** address lines go straight to the chip (13 lines for an 8 KB chip). The **high** lines go to a **decoder** that drives the chip's $\overline{CS}$, so each chip answers only in its own address block.

Full decoding: every high line is decoded → each location has **exactly one** address. Needed for a clean, expandable memory map.

Partial decoding: some high lines ignored → cheaper, but each location repeats at **2^k addresses** (k = ignored lines): **fold-back / aliasing / mirror images.**

The 74LS138 3-to-8 decoder



C = MSB; enabled by G1 = 1, $\overline{G2A}$ = $\overline{G2B}$ = 0

Output	A15 A14 A13	Address range
$\overline{Y0}$	0 0 0	F0000H–F1FFFH
$\overline{Y1}$	0 0 1	F2000H–F3FFFH
$\overline{Y2}$	0 1 0	F4000H–F5FFFH
$\overline{Y3}$	0 1 1	F6000H–F7FFFH
$\overline{Y4}$	1 0 0	F8000H–F9FFFH
$\overline{Y5}$	1 0 1	FA000H–FBFFFH
$\overline{Y6}$	1 1 0	FC000H–FDFFFH
$\overline{Y7}$	1 1 1	FE000H–FFFFFH

Worked example: memory map ✍️: 8088, eight 2764 EPROMs (8K × 8) filling F0000H–FFFFFH.

1. 8K = 2^{13} → A0–A12 to every chip
2. A13–A15 → C B A inputs pick 1 of 8 chips
3. A16–A19 must be 1111 → NAND → $\overline{G2A}$; IO/$\overline{M}$ = 0 for memory → $\overline{G2B}$
4. 13 + 3 + 4 = 20 lines used → **full decoding**

Formulas:

End = Start + Size − 1

8 KB = 2000H: F0000H + 1FFFH = F1FFFH

F5A3CH: 5A3CH = **010**1 1010 … → $\overline{Y2}$

Partial decoding ✍️: A 2K RAM (6116) with $\overline{CS}$ = A19 only: A0–A10 on chip, A11–A18 ignored (8 lines) → **2^8 = 256 images:** 00000–007FFH, 00800–00FFFH … 7F800–7FFFFH.

Memory trick: "**Size picks the low lines, the decoder gets the rest.**" Lines on chip + decoded lines = 20 → full; fewer → partial.

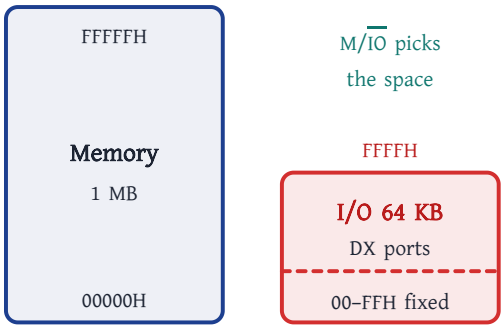
Exam trap: Images = 2^k, not 2k. Boot ROM must cover **FFFF0H** (reset vector), so EPROM goes at the top, RAM from 00000H (IVT).

9 Isolated (I/O-mapped) vs Memory-mapped I/O

	Isolated / port-mapped I/O	Memory-mapped I/O
Space	separate: **64K byte ports**	inside the 1 MB (uses up addresses)
Instructions	**IN / OUT** only	any memory instruction (MOV, AND ...)
Data via	AL or AX only	any register
Signals	M/$\overline{IO}$ = 0; $\overline{IORC}$, $\overline{IOWC}$	M/$\overline{IO}$ = 1; $\overline{MRDC}$, $\overline{MWTC}$
Decoding	8 or 16 lines (simpler)	all 20 lines
Typical	x86 PCs (Intel)	68000, ARM, microcontrollers

```
IN  AL, 60H    ; fixed port 00H-FFH
IN  AX, DX     ; DX = 0000H-FFFFH
OUT 43H, AL    ; byte to port 43H
MOV DX, 0378H  ; port above FFH:
OUT DX, AL     ; must use DX
```

Fixed port → **256 ports** (A8–A15 = 00H). DX → **65,536 ports**. A16–A19 = 0 in every I/O cycle.



two separate address spaces

Exam trap: **IN BL, 60H** is illegal (only AL/AX); **OUT 378H, AL** is illegal (port > FFH needs DX). Memory-mapped I/O has **no** IN/OUT cycles at all.

10 I/O Port Decoding & Types of I/O

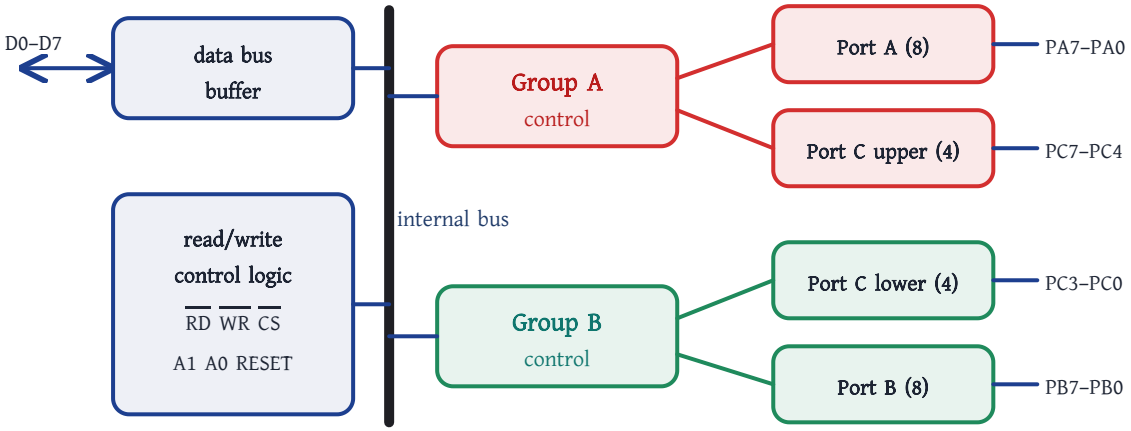
Decoding ✍️: Ports F8H–FFH on an 8088: A7–A3 = 11111 → 74LS30 NAND → $\overline{G2A}$; IO/$\overline{M}$ (1 for I/O) → G1; A2 A1 A0 → C B A, so $\overline{Yn}$ = port F8H + n. A8–A15 are ignored (partial decoding for DX ports). Input port: a **74LS244** enabled only by its $\overline{IORC}$; output port: a **74LS374** latch clocked by $\overline{IOWC}$.

Method	How	Verdict
Programmed (polling)	CPU loops on a status bit	simple, wastes CPU time
Interrupt-driven	device raises INTR (8259A, mp4)	CPU free until needed
DMA	8237 takes the bus (HOLD/HLDA)	fastest for blocks

IT link: The IBM PC decodes only A0–A9 (000H–3FFH): 20H 8259A · 40H–43H 8254 · **60H–63H 8255** (PC/XT) · 378H LPT1 · 3F8H COM1. Parallel vs serial = number of data lines, not a transfer method (serial 8251: mp3).

11 8255A Programmable Peripheral Interface

Definition: A 40-pin chip with **24 I/O lines** in three 8-bit ports A, B, C. **Group A** = Port A + PC7–PC4 (upper C); **Group B** = Port B + PC3–PC0 (lower C). A control word sets each port's direction and mode.



8255A block diagram: red = Group A, green = Group B

A1 A0	Selects	8088 / 8086
0 0	Port A	60H / 80H
0 1	Port B	61H / 82H
1 0	Port C	62H / 84H
1 1	**Control register**	63H / 86H

($\overline{CS}$ = 0.) On an 8086 the 8255A sits on D0–D7, so CPU **A1, A2** drive its A0, A1 → **even port addresses** (80H, 82H, 84H, 86H).

RESET → all 24 lines become **inputs** (mode 0), so nothing is driven by accident at power-on.

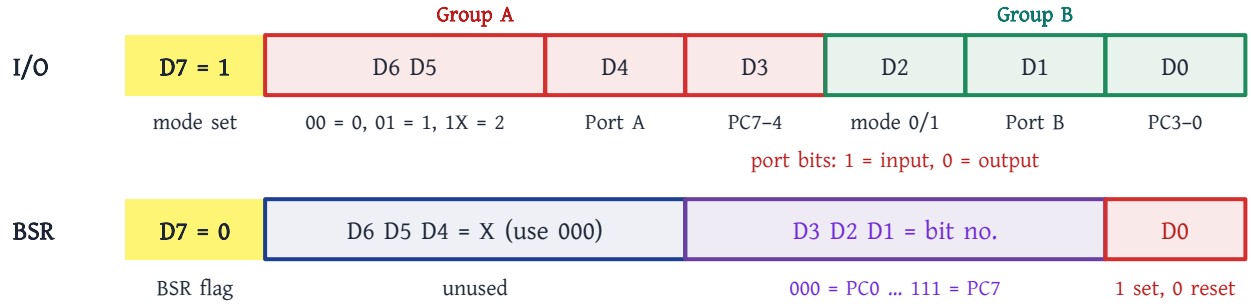
12 The Three Modes (+ BSR)

Mode	Name	Ports	Key points
0	Basic I/O	A, B, C upper, C lower	no handshake; outputs latched, inputs not; 16 in/out combinations
1	Strobed I/O	A and B	handshake on Port C: A uses PC3–PC5 (in) or PC3, PC6, PC7 (out); B uses PC0–PC2
2	Bidirectional	**Port A only**	PC3–PC7 = INTR, $\overline{STB}$, IBF, $\overline{ACK}$, $\overline{OBF}$
BSR	Bit set/reset	Port C	sets or clears one PC bit at a time

Memory trick: "**A-B-C = 3-2-1**": Port A has 3 modes (0, 1, 2), Port B 2 (0, 1), Port C 1 (mode 0) plus BSR.

Exam trap: Port C has no handshake mode of its own; in modes 1/2 some of its lines become handshake pins and the **leftover** lines stay normal I/O.

13 8255A Control Words: I/O Mode & BSR Mode



D7 picks the format: 1 = mode-definition word, 0 = bit set/reset word

Required set-up	D7 D6D5 D4 D3 D2 D1 D0	Word
all ports output (mode 0)	1 00 0 0 0 0 0	80H
all ports input (mode 0)	1 00 1 1 0 1 1	9BH
A in, C_U out, B out, C_L in	1 00 1 0 0 0 1	91H
A out, C_U in, B in, C_L out	1 00 0 1 0 1 0	8AH
A mode 1 in, B mode 0 out	1 01 1 0 0 0 0	B0H
BSR: set PC5 · reset PC5	0 000 101 1 · 0 000 101 0	0BH · 0AH

```
MOV AL, 91H  ; A in, B out
OUT 86H, AL  ; control (8086)
IN  AL, 80H  ; read Port A
OUT 82H, AL  ; write Port B
MOV AL, 0BH  ; BSR: set PC5
OUT 86H, AL  ; also to CONTROL
```

Memory trick: BSR word for PCn = **2n + 1** to set, **2n** to reset (PC7: 0FH / 0EH).

Exam trap: BSR words go to the **control register**, never to Port C. A new mode word **clears all output latches**.

Interface examples (syllabus list)

Device	Hook-up · key fact
Keyboard matrix	rows ← output port, columns → input port; scan, **debounce 10–20 ms**
7-segment	common cathode: 0 → 3FH, 1 → 06H, 8 → 7FH; anode = complement
LCD 16 × 2	RS = 0 command, 1 data; E latches; 38H, 0EH, 01H (clear), 80H
Printer	Port A mode 1 out; printer $\overline{\text{ACK}}$ → 8255A $\overline{\text{ACK}}$; $\overline{\text{STROBE}}$ via BSR; poll BUSY
Stepper motor	full step 09H → 0CH → 06H → 03H; 1.8° = 200 steps/rev
ADC / DAC 0808	8-bit: step 5 V/256 = **19.5 mV**; DAC 80H → 2.5 V

Night-Before-the-Exam Sheet

Fact sheet:

CLK = f_{xtal} ÷ 3 (33%) · PCLK = CLK ÷ 2

15 MHz → 5 MHz · 24 MHz → 8 MHz

Bus cycle = 4T; +1 T per wait state

5 MHz: T = 200 ns, cycle = 800 ns

MN/$\overline{MX}$ = 1 min · 0 max (8288)

8288: $\overline{S2}$ = 0 I/O, 1 memory

ALE → 373 latch · $\overline{DEN}$, DT/$\overline{R}$ → 245

8086: 3 × 373, 2 × 245 · 8088: 1 × 245

N locations → $\log_2 N$ lines (8K → 13)

Images = 2^k (k = ignored lines)

$\overline{BHE}$ A0: 00 word · 01 odd · 10 even

Reset: CS:IP = FFFF:0000 → FFFF0H

IN/OUT: AL/AX; fixed 00–FFH; DX 0–FFFFH

8255: A1A0 = 11 → control register

CW: D7 = 1 mode set, 0 BSR · 1 = input

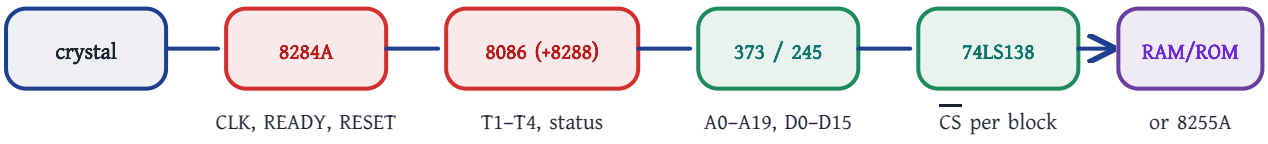
80H all out · 9BH all in · 91H, 8AH

Top 12 one-liners for MCQs

1. 8284A divides the crystal by 3
2. 8288 is used in maximum mode only
3. Bus arbiter = 8289 (not the 8288)
4. 74LS373 = address latch, strobed by ALE
5. 74LS245 = bidirectional data transceiver
6. 74LS138: 3-to-8, active-low outputs
7. 32K × 8 chip: 15 address, 8 data lines
8. 8255A: 3 ports, 24 lines, 40 pins
9. Mode 2 (bidirectional): Port A only
10. Port B: modes 0 and 1 only
11. RESET makes all 8255A ports inputs
12. Isolated I/O: 64K ports, IN/OUT

Top traps: CLK duty 33%, not 50% · wait state = READY low · 8288 DEN is active high · images = 2^k, not 2k · IN/OUT only via AL/AX; ports > FFH need DX · BSR words go to the control register · Port C has no mode 1/2 of its own

All tricks together: OSC × 1, CLK ÷ 3, PCLK ÷ 6 · $\overline{S2}$ = 0 is the I/O side · ALE → Address, DEN → Data, DT/$\overline{R}$ → Direction · 373 latch, 244 one-way, 245 two-way · chip name digits = Kbits · size picks the low lines, the decoder gets the rest · A-B-C = 3-2-1 · BSR = 2n + 1 (set), 2n (reset)



the signal path of a whole 8086 board, left to right

Written-exam tips: Memory design: chip lines → decoder wiring (C B A + enables) → **memory map table** (start/end) → full or partial. 8255A: write the word bit by bit, then the hex.

✓ Practice MCQs (exam pattern, 10 marks)

1. 8284A + 24 MHz crystal → CLK =
(a) 12 MHz (b) 24 MHz
(c) 8 MHz (d) 6 MHz

2. The 8288 is needed in the 8086's:
(a) min mode (b) single step
(c) DMA mode (d) max mode

3. The 74LS373 latches the address on:
(a) $\overline{\text{DEN}}$ (b) ALE
(c) DT/$\overline{\text{R}}$ (d) READY

4. Address lines of an 8K × 8 chip:
(a) 13 (b) 8
(c) 12 (d) 16

5. 2K RAM, $\overline{\text{CS}}$ = A19 only: images?
(a) 2 (b) 8
(c) 256 (d) 512

6. 8255A BSR word to set PC6:
(a) 0DH (b) 0CH
(c) 86H (d) 06H

7. Max. 8-bit ports reachable via DX:
(a) 1024 (b) 256
(c) 1 M (d) 65,536

8. 8255A mode 2 is allowed on:
(a) Port B (b) Port C
(c) Port A (d) Ports A, B

9. 5 MHz, one wait state: bus cycle =
(a) 800 ns (b) 1000 ns
(c) 600 ns (d) 1200 ns

10. $\overline{\text{BHE}}$ = 0, A0 = 1 transfers:
(a) even word (b) nothing
(c) even byte (d) odd byte

Answer key → 1-c · 2-d · 3-b · 4-a · 5-c · 6-a · 7-d · 8-c · 9-b · 10-d

Written question: An 8255A sits at ports 40H–43H of an 8088. Program Port A as input, Port B as output, upper C as output, lower C as input (mode 0); copy the switches on Port A to the LEDs on Port B; then set PC6 with a BSR command. 8 marks

1. 8255A A1 A0 = CPU A1 A0 → PA 40H, PB 41H, PC 42H, control **43H** 1
2. D7 = 1, mode A 00, A in 1, C_U out 0, mode B 0, B out 0, C_L in 1 → 1001 0001 = **91H** 2
3. Code (right): 91H → 43H, IN from 40H, OUT to 41H 3
4. BSR: 0 000 110 1 = **0DH**, sent to the **control port 43H**, not Port C 2

```
MOV AL, 91H
OUT 43H, AL
IN  AL, 40H
OUT 41H, AL
MOV AL, 0DH
OUT 43H, AL
```

Before the exam ✓

- ✓ Can draw the 8284A, 373, 245 hook-up
- ✓ Can draw T1–T4 with ALE, $\overline{\text{RD}}$, READY
- ✓ Know the 8288 status table
- ✓ Can build a 74LS138 memory map
- ✓ Can write any 8255A mode/BSR word
- ✓ Know IN/OUT rules (AL/AX, DX)